



HCMUTE



fie.hcmute.edu.vn

GRADUATION PROJECT

DEVELOPMENT OF A MULTIMODAL 3D OBJECT RECOGNITION AND VISUALIZATION SYSTEM FOR TRAFFIC ENVIRONMENTS USING CAMERA AND LIDAR FUSION

Lê Hoàng Tiến

21145617

Lê Hồ Minh Khoa

21145019

Supervisor

Trần Vũ Hoàng, PhD.

TABLE OF CONTENT

1

Introduction

2

Proposed Pipeline

Input & Setup Data

Model Integration & Inference

Evaluation & Visualization

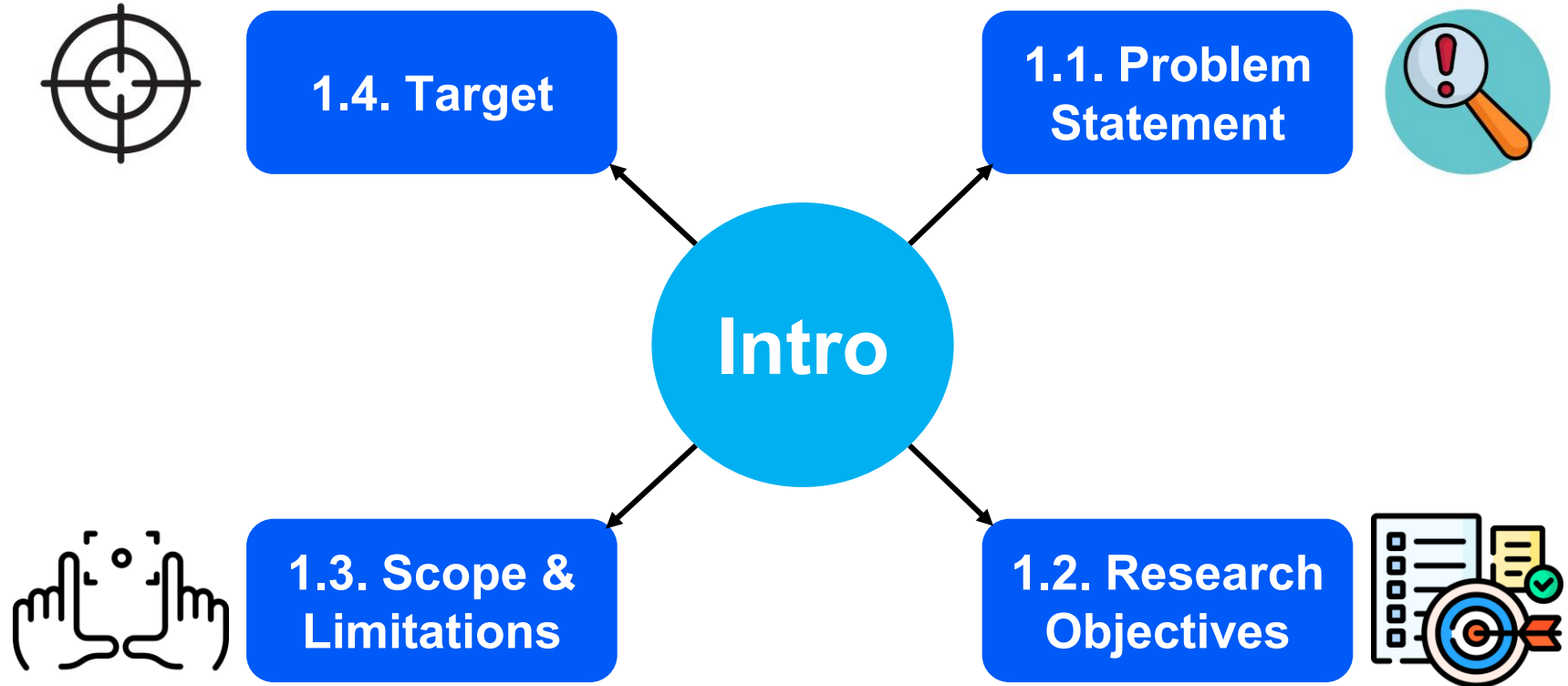
3

Results

4

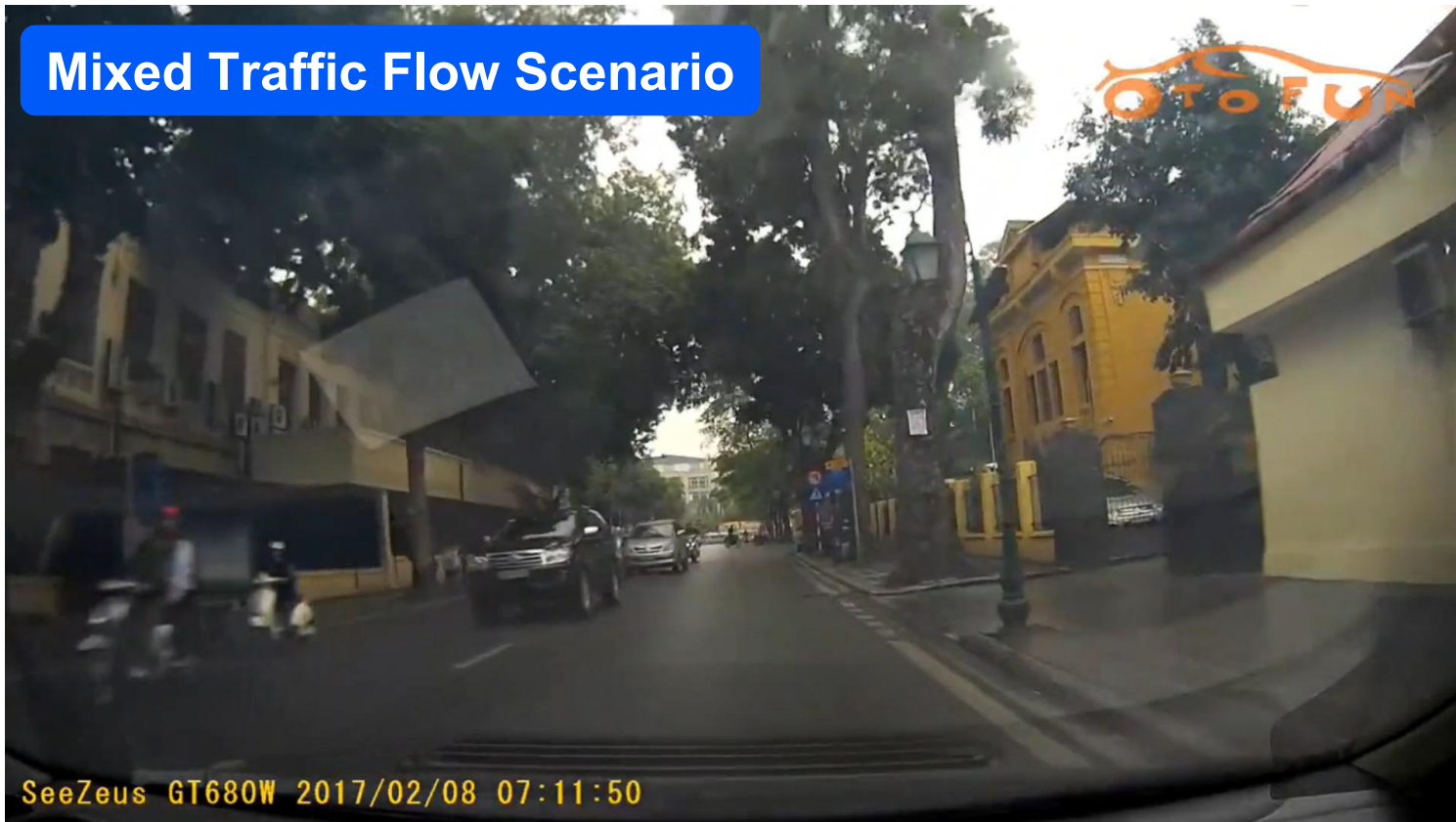
Conclusion

1. INTRODUCTION



1.1. PROBLEM STATEMENT

Mixed Traffic Flow Scenario



1.1. PROBLEM STATEMENT

Low Lighting Scenario

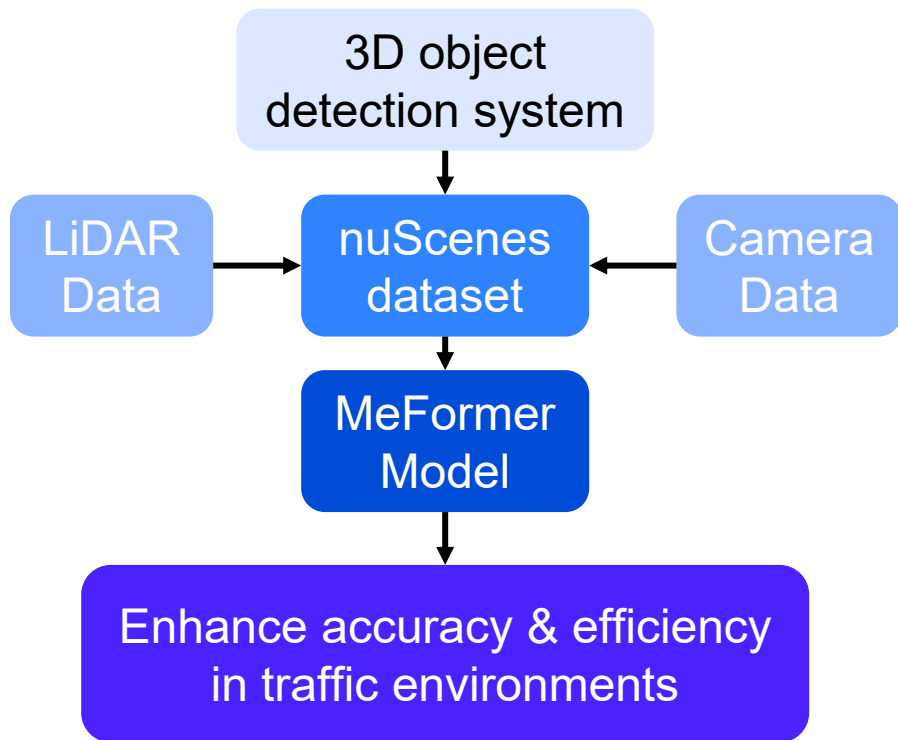
Warning

Some viewers may find the following footage distressing

**The
Guardian**

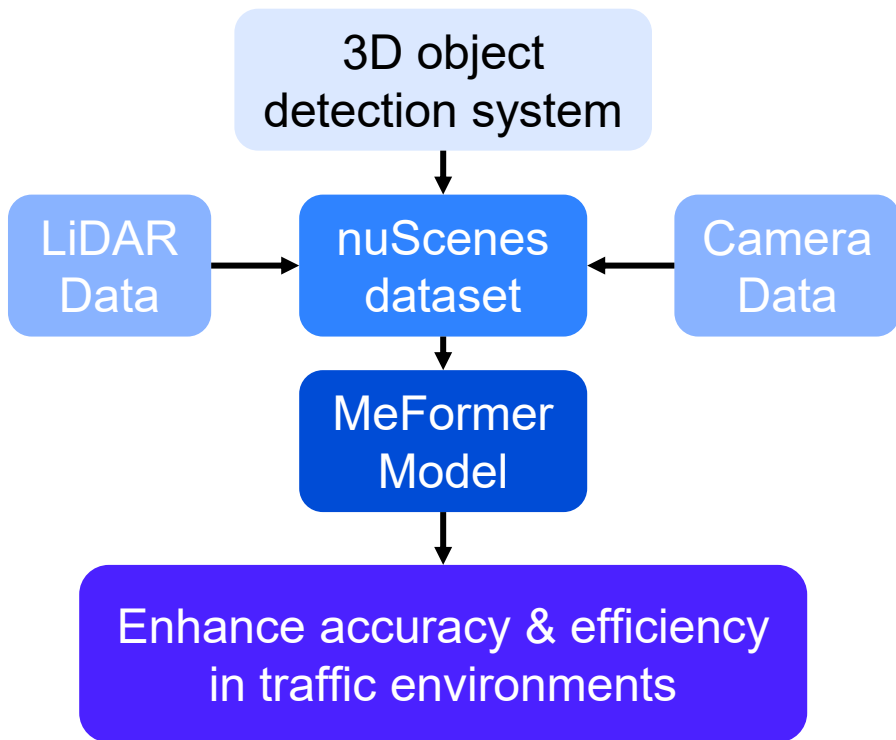
1. INTRODUCTION

1.2. RESEARCH OBJECTIVES

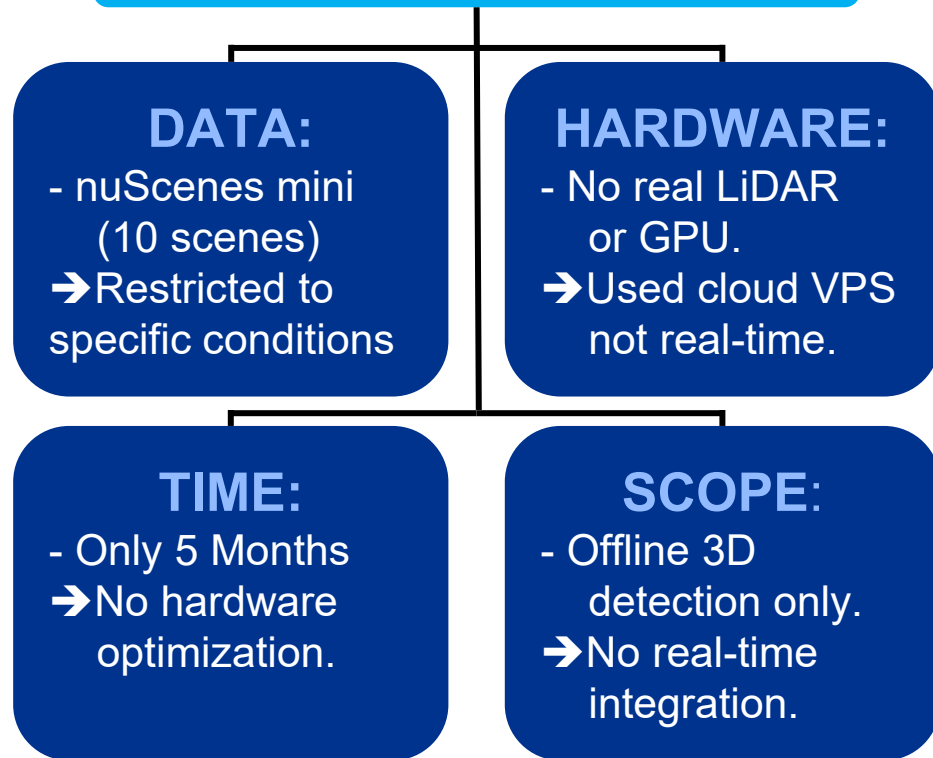


1. INTRODUCTION

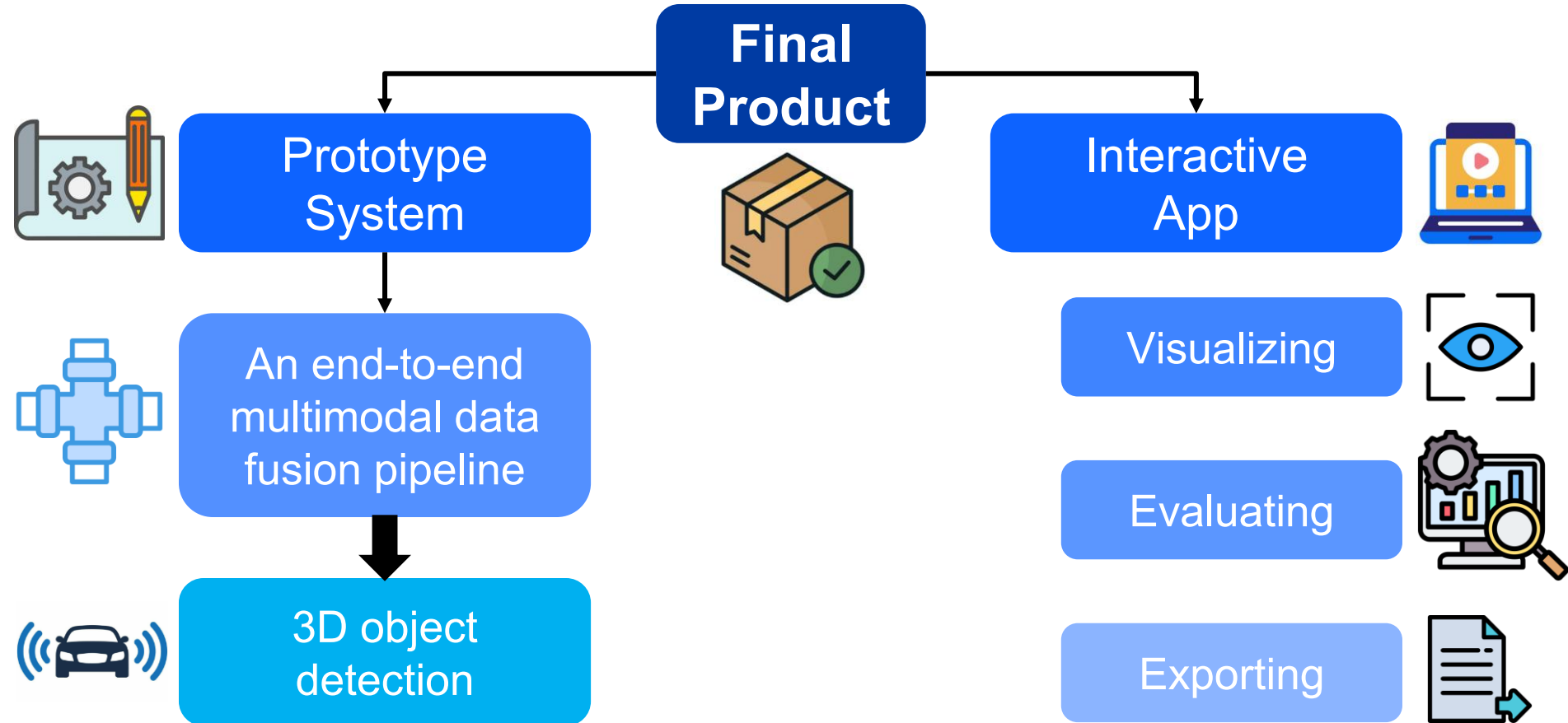
1.2. RESEARCH OBJECTIVES



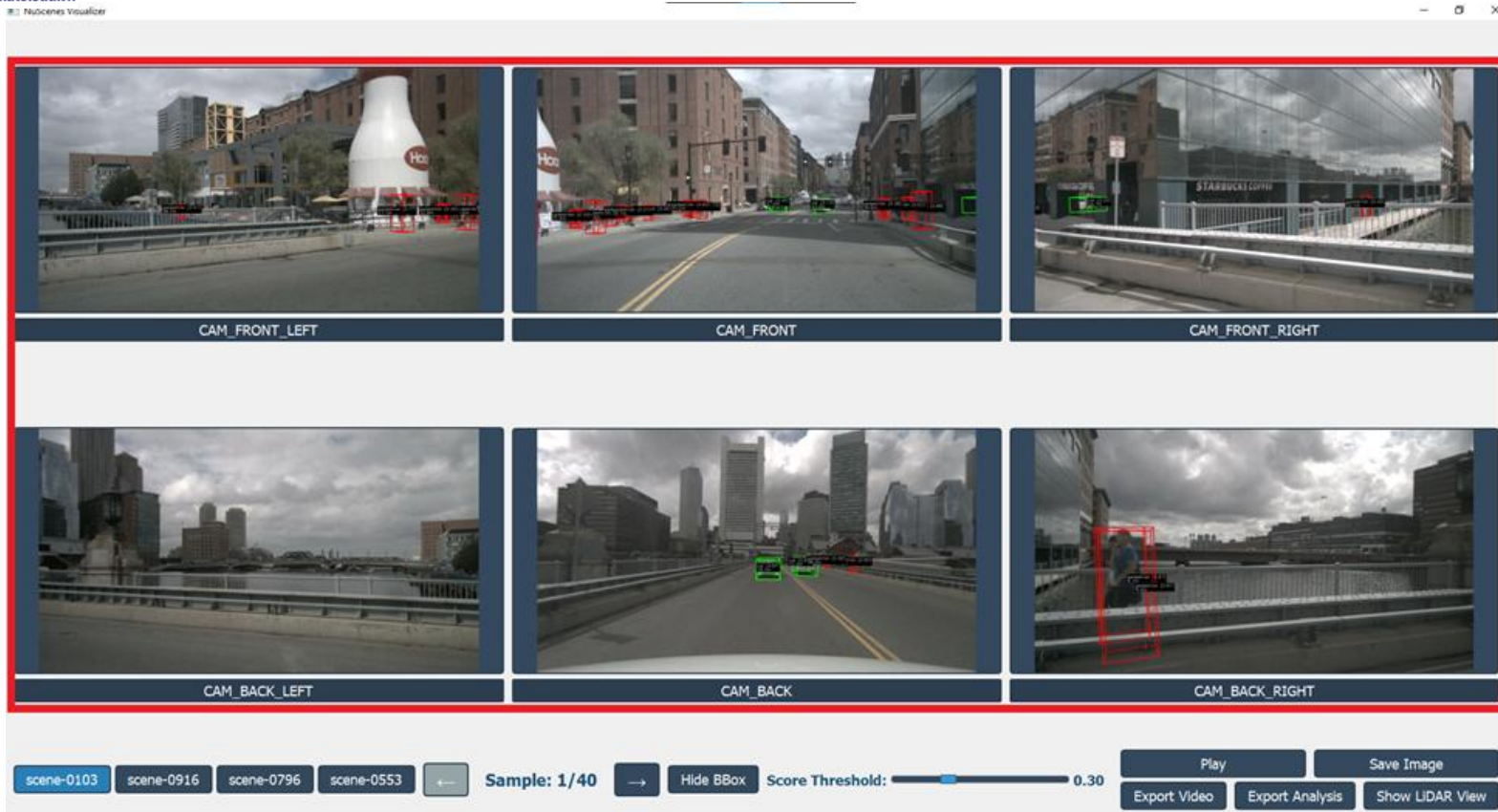
1.3. SCOPE & LIMITATIONS



1.4. TARGET

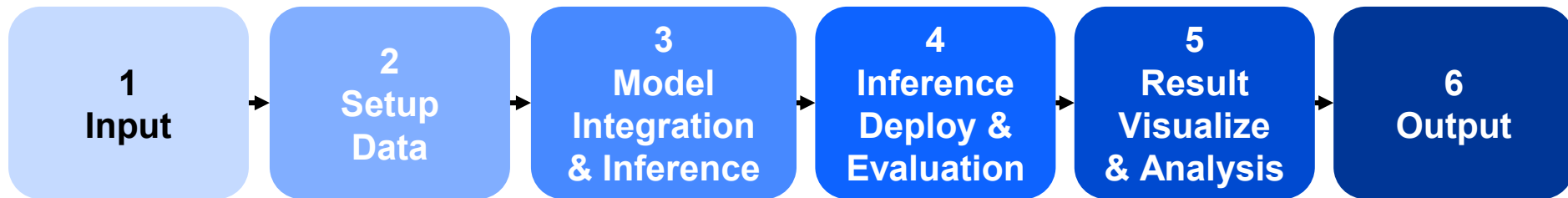


1.4. TARGET



Main User Interface of the Application

2. OVERVIEW OF PIPELINE



2. OVERVIEW OF PIPELINE

- Camera images
- LiDAR point cloud
- Calibration parameters

1
Input

2
**Setup
Data**

3
**Model
Integration
& Inference**

4
**Inference
Deploy &
Evaluation**

5
**Result
Visualize
& Analysis**

6
Output

2. OVERVIEW OF PIPELINE

- Camera images
- LiDAR point cloud
- Calibration parameters

1
Input

2
Setup
Data

3
Model
Integration
& Inference

4
Inference
Deploy &
Evaluation

5
Result
Visualize
& Analysis

6
Output

- Standardize the format for MEFormer
- Integrate images, point clouds & calibration

2. OVERVIEW OF PIPELINE

- Camera images
- LiDAR point cloud
- Calibration parameters

- Detect 3D bounding boxes & labels with MEFormer
- Apply Non-Max Suppression & filter results

2
Setup
Data

3
Model
Integration
& Inference

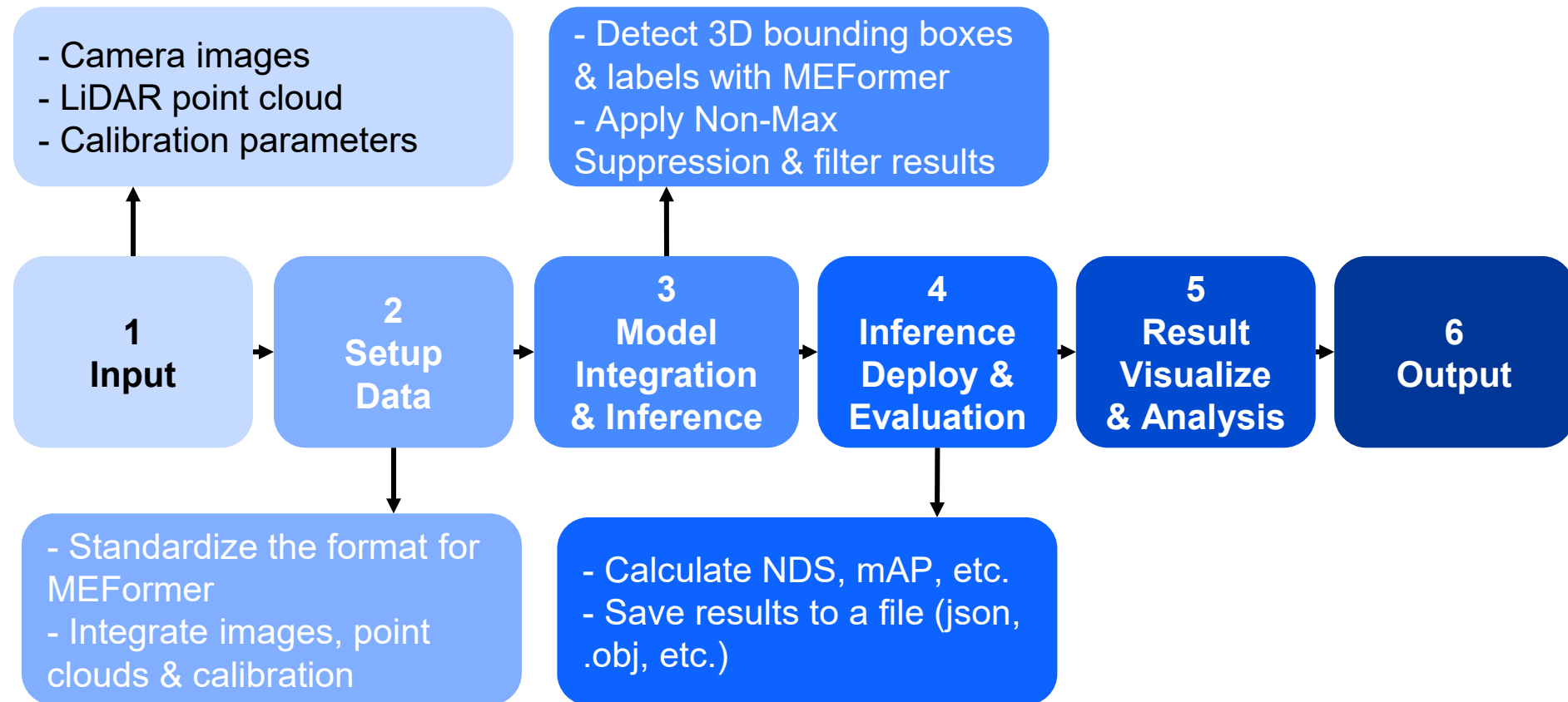
4
Inference
Deploy &
Evaluation

5
Result
Visualize
& Analysis

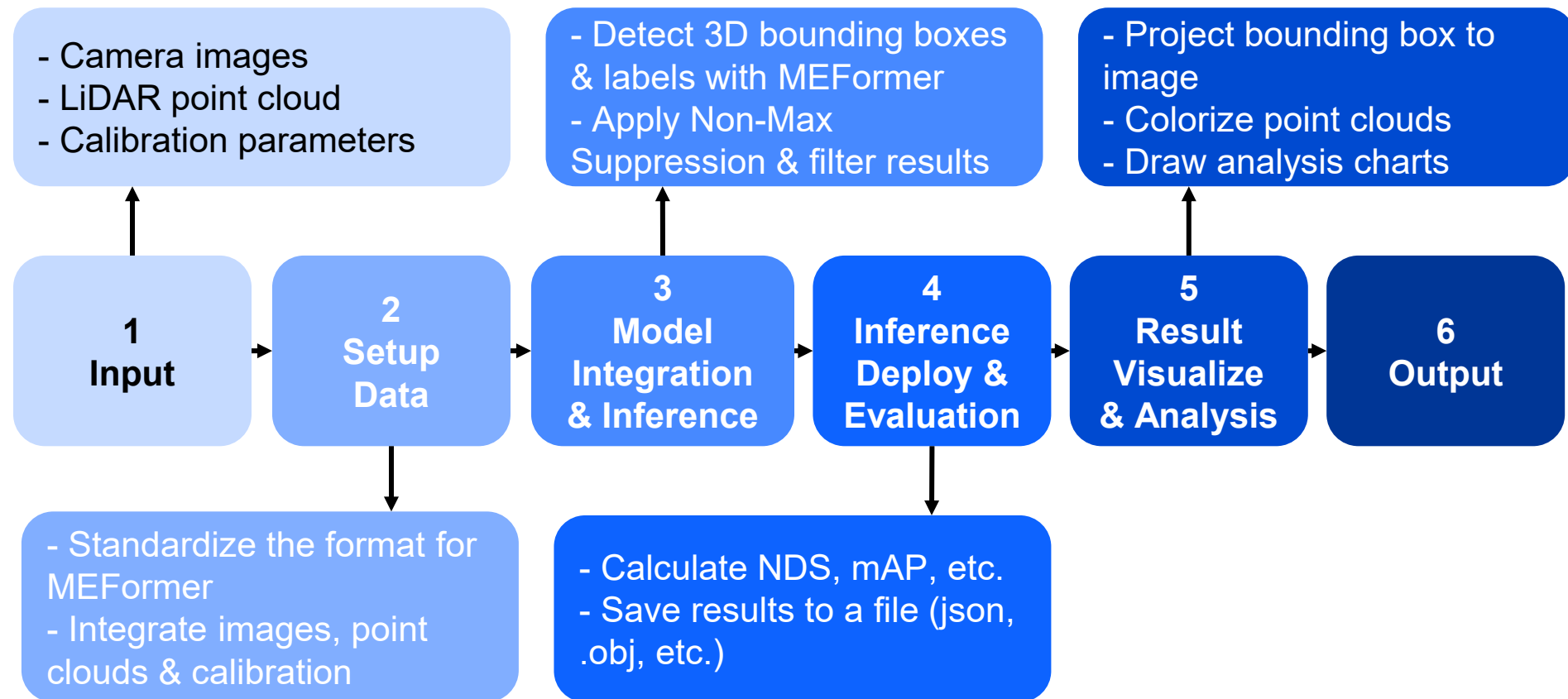
6
Output

- Standardize the format for MEFormer
- Integrate images, point clouds & calibration

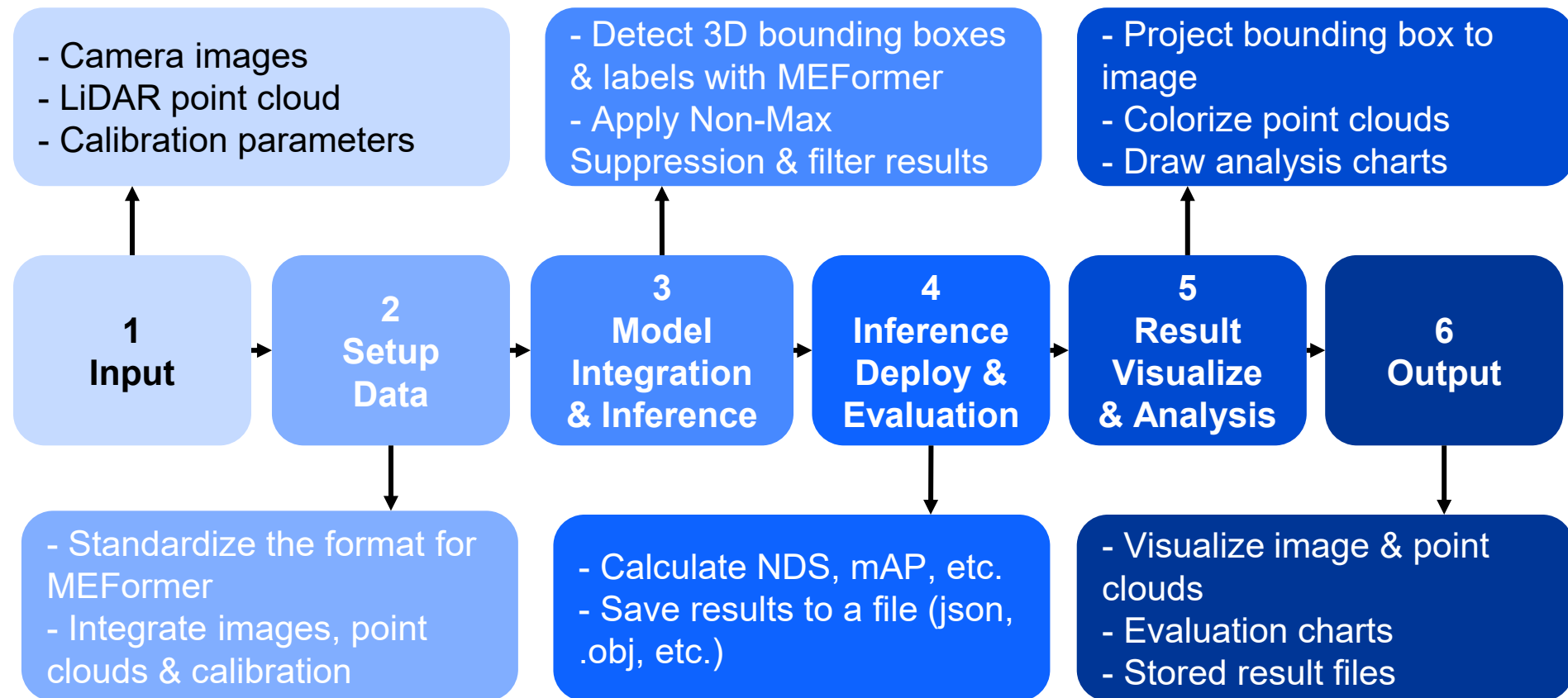
2. OVERVIEW OF PIPELINE



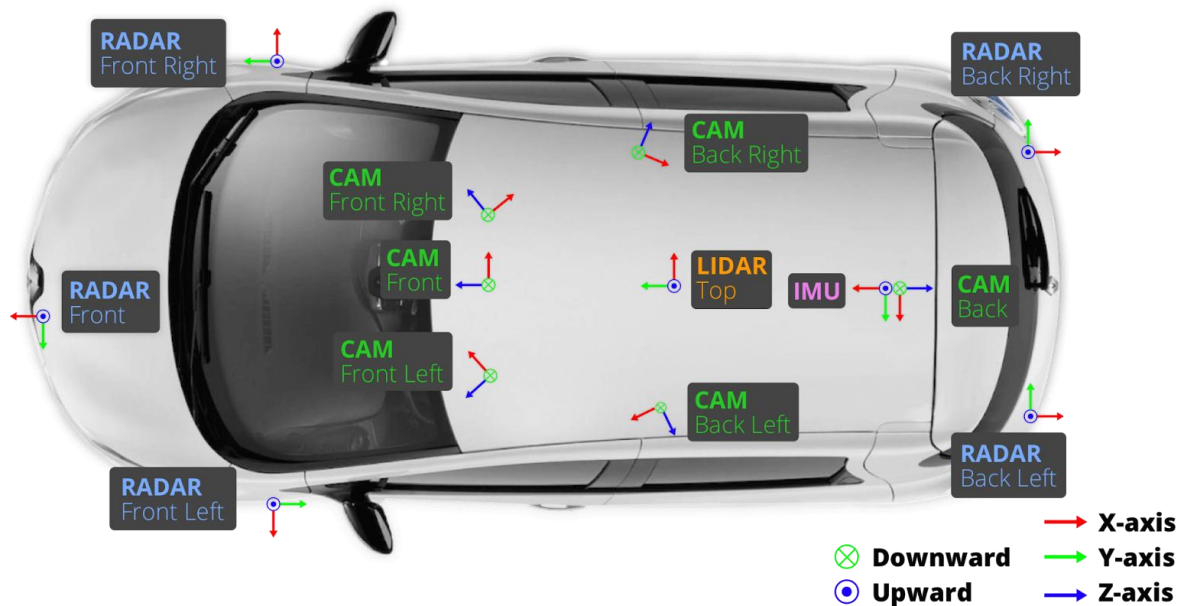
2. OVERVIEW OF PIPELINE



2. OVERVIEW OF PIPELINE



2.1. INPUT & SETUP DATA



Autonomous Vehicle Sensor Layout
in nuScenes Dataset

6 CAMERAS

- Front
- Front-left
- Front-right
- Rear
- Rear-left
- Rear-right

1 LIDAR

- A 32-channel device

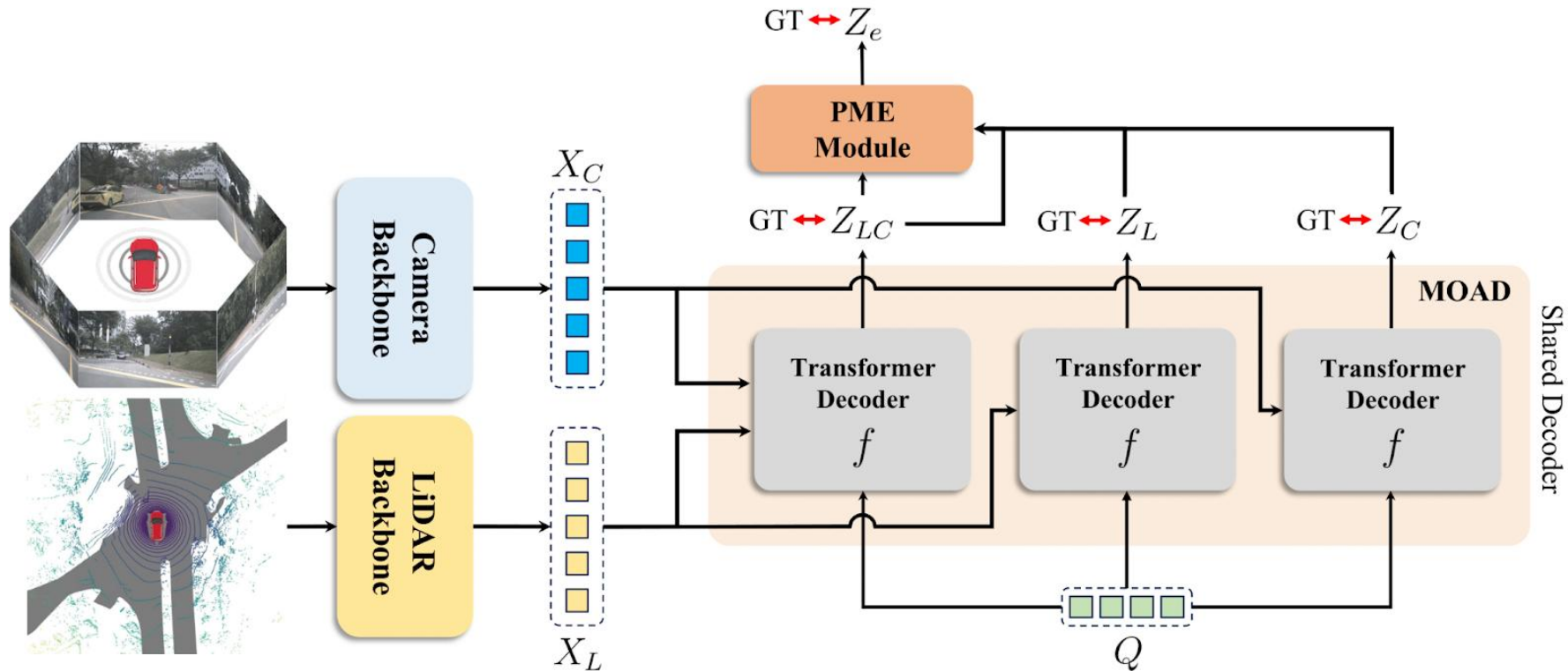
5 RADARS

- Front & 4 Edges

1 IMU

- On the roof

2.2. MODEL INTEGRATION & INFERENCE



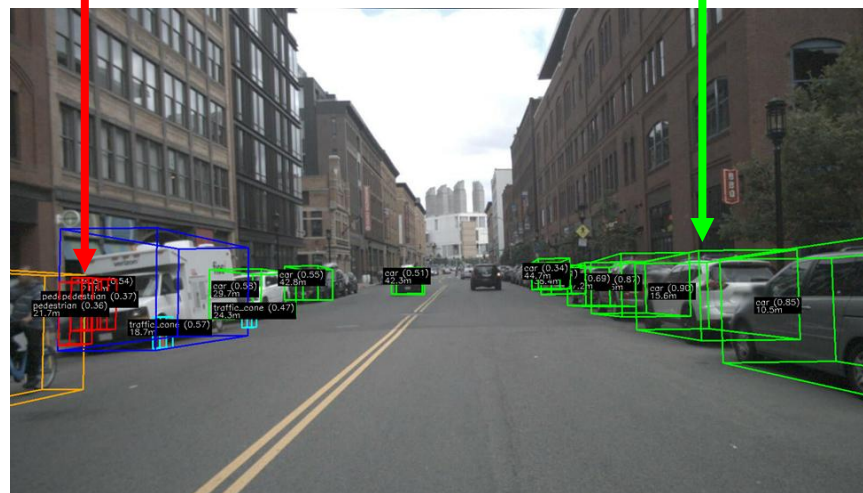
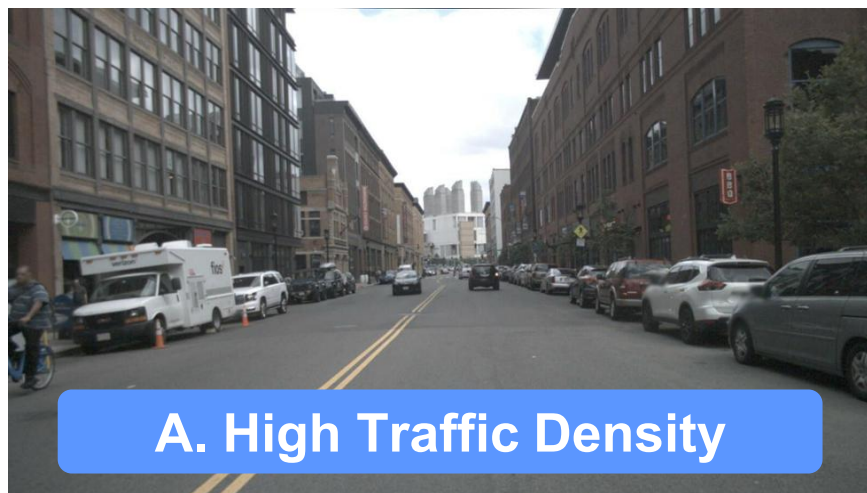
The overall architecture of MEFormer

2.2. MODEL INTEGRATION & INFERENCE

MODELS	mAP (%)	mATE (m)	mAOE (rad)
BEVFormer (C)	44.5	0.58	0.38
DETR3D (C)	41	0.64	0.39
BEVFusion (L+C)	71.3	0.25	0.36
MEFormer (L+C)	72	0.27	0.3
Expected Metrics	≥ 53	≤ 0.49	≤ 0.38
Result	74	0.27	0.21

Performance Comparison of 3D Object Detection Models

2.3. EVALUATION AND VISUALIZATION



Scene-0103, Sample 34 with threshold > 0.3

2.3. EVALUATION AND VISUALIZATION



Scene-0916, Sample 8 with threshold > 0.3

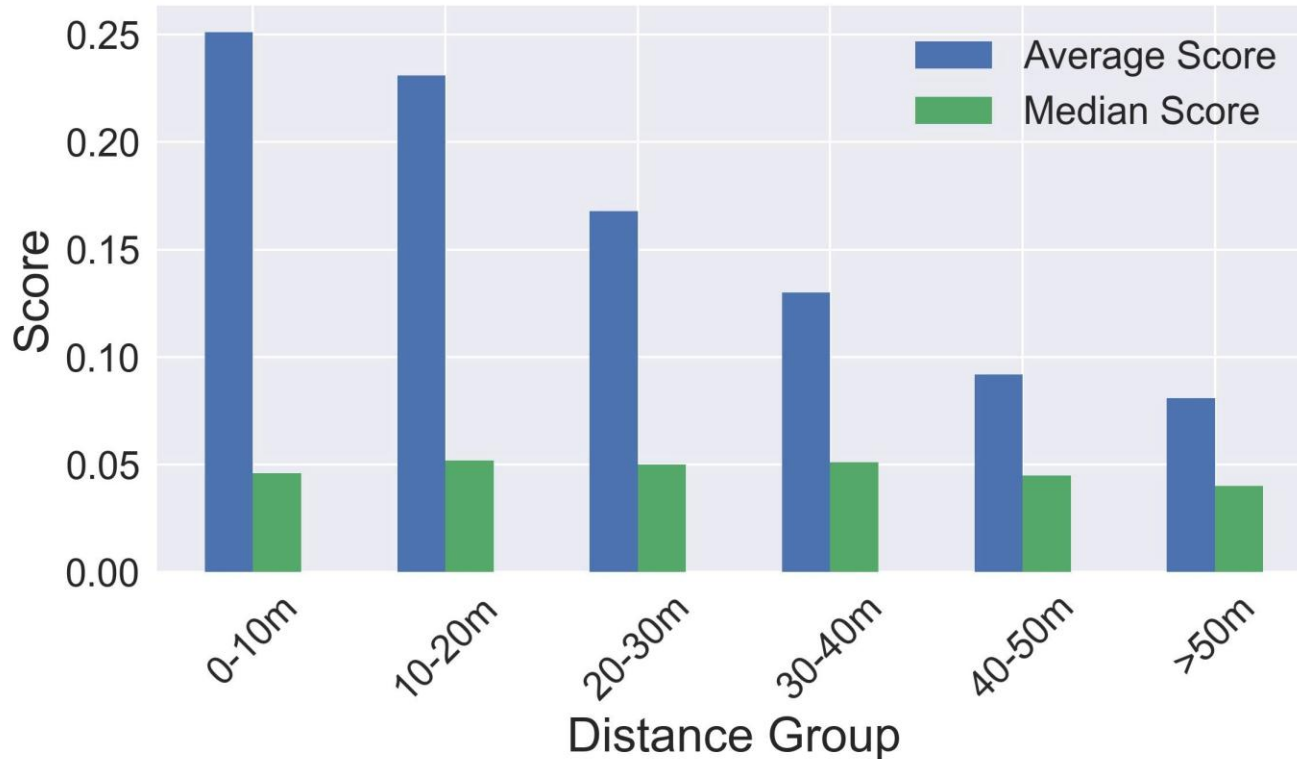
2.3. EVALUATION AND VISUALIZATION



Scene-0796, Sample 11 with threshold > 0.3

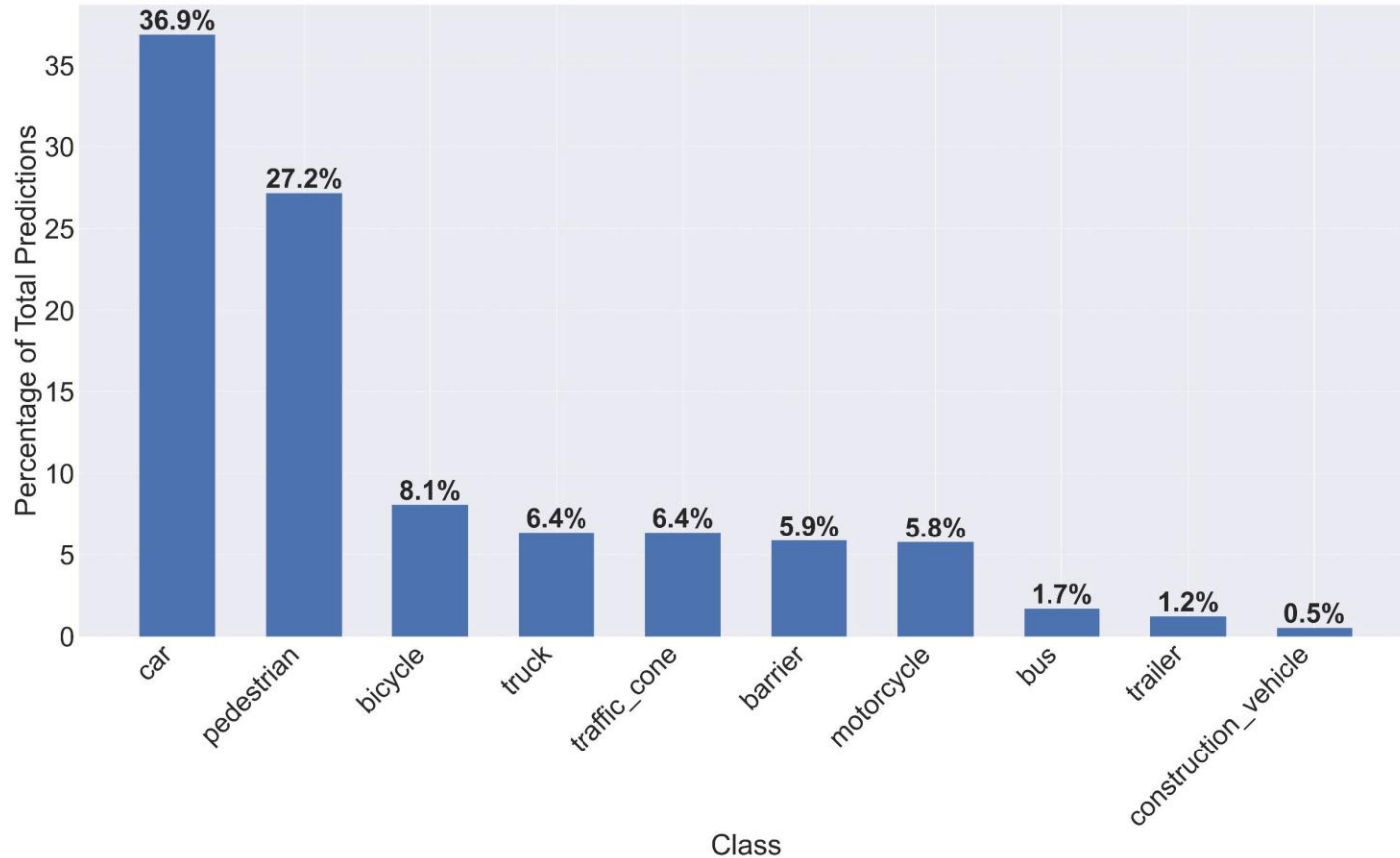
2.3. EVALUATION AND VISUALIZATION

Average and Median Scores by Distance Group



2.3. EVALUATION AND VISUALIZATION

Distribution of Predictions by Class (%)



THE 3-SECOND RULE

A traffic safety guideline for drivers



~2.5-second reaction distance
at 50 km/h (~13.9 m/s)



Recommended Distance = $2.5s \times 13.9m/s = 34.75m$

Our MEFormer model is most reliable under 40 meters



Our MEFormer model meets the 3-Second Rule



DEMO PRODUCT

3. CONCLUSION

Summary

Implemented MeFormer with
MMDetection3D on nuScenes.

1

2

Achieved high mAP & NDS,
good for car, bus, pedestrian.

3

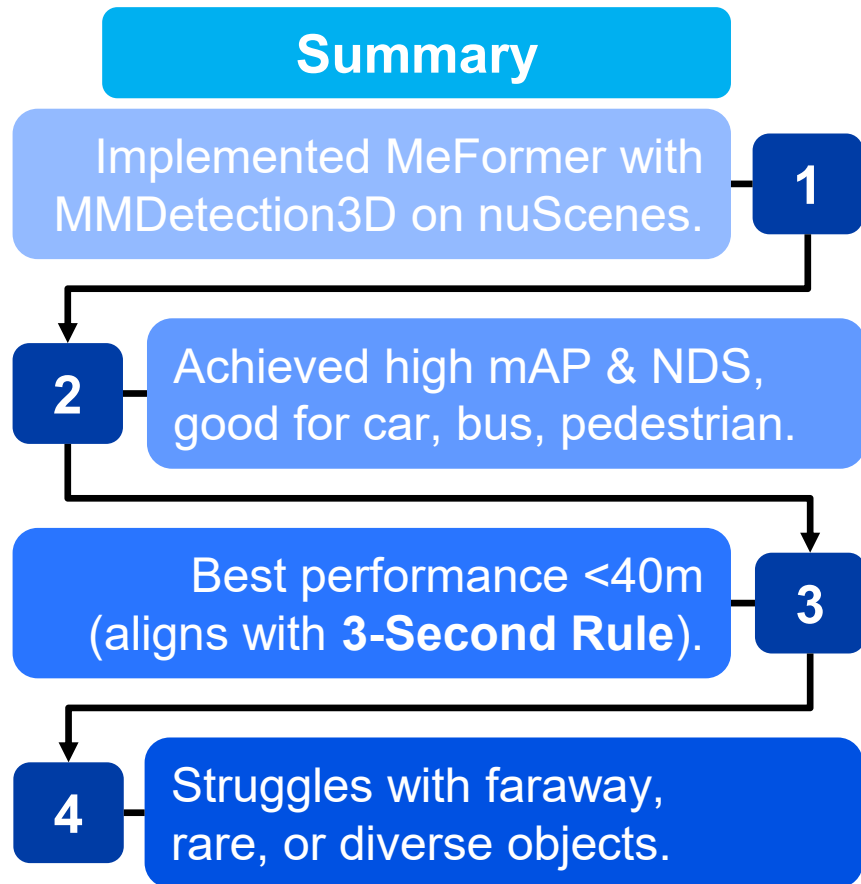
Best performance <40m
(aligns with **3-Second Rule**).

4

Struggles with faraway,
rare, or diverse objects.

3. CONCLUSION

Summary



Future Development

- A HARDWARE SYSTEM:** Build LiDAR/camera system for collecting real-world data.
- B REAL-WORLD TEST:** Deploy on vehicles/simulators to check performance & latency.
- C RARE CLASS BOOST:** Use augmentation/transfer learning for rare/small classes.
- D DYNAMIC THRESHOLDS:** Tune score thresholds by distance or application.



HCMUTE



fie.hcmute.edu.vn

THANK YOU!

**Development of a multimodal 3D object
recognition and visualization system for traffic
environments using camera and LiDAR fusion**

GRADUATION PROJECT